OmniTrace: A Unified Framework for Generation-Time Attribution in Omni-Modal LLMs

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Modern multimodal large language models (MLLMs) generate fluent responses from interleaved text, image, audio, and video inputs. However, identifying which input sources support each generated statement remains an open challenge. Existing attribution methods are primarily designed for classification settings, fixed prediction targets, or single-modality architectures, and do not naturally extend to autoregressive, decoder-only models performing open-ended multimodal generation. We introduce OmniTrace, a lightweight and model-agnostic framework that formalizes attribution as a generation-time tracing problem over the causal decoding process. OmniTrace provides a unified protocol that converts arbitrary token-level signals such as attention weights or gradient-based scores into coherent span-level, cross-modal explanations during decoding. It traces each generated token to multimodal inputs, aggregates signals into semantically meaningful spans, and selects concise supporting sources through confidence-weighted and temporally coherent aggregation, without retraining or supervision. Evaluations on Qwen2.5-Omni and MiniCPM-o-4.5 across visual, audio, and video tasks demonstrate that generation-aware span-level attribution produces more stable and interpretable explanations than naive self-attribution and embedding-based baselines, while remaining robust across multiple underlying attribution signals. Our results suggest that treating attribution as a structured generation-time tracing problem provides a scalable foundation for transparency in omni-modal language models.

Project Page: https://github.com/eric-ai-lab/OmniTrace

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1. Introduction

Multimodal large language models (MLLMs) are increasingly deployed in settings that require not only fluent generation but also transparent grounding in heterogeneous input evidence, including text, images, audio, video and their combinations [1, 2, 3, 4, 5, 6, 7, 8]. In multimodal summarization, reasoning, and decision support, models must not only generate fluent outputs but also justify which input segments: text spans, image regions, audio and video intervals support each statement. This requirement has renewed interest in attribution methods that can trace model outputs back to their supporting sources.

Prior work on neural attribution has largely focused on classification settings and encoder-based architectures, where explanations are computed with respect to a fixed objective such as a class logit or an extractive span [9, 10, 11, 12, 13, 14, 15, 16, 17]. These approaches typically produce token-level importance scores derived from attention weights, gradients, or relevance propagation. While effective in constrained settings, they do not directly address the attribution problem posed by contemporary decoder-only MLLMs. In open-ended generation, there is no externally specified target text, and attribution must operate over a growing causal graph whose structure evolves at each decoding step. Moreover, modern models operate over interleaved multimodal inputs, where attribution must span text, image, audio and video tokens within a unified causal sequence.

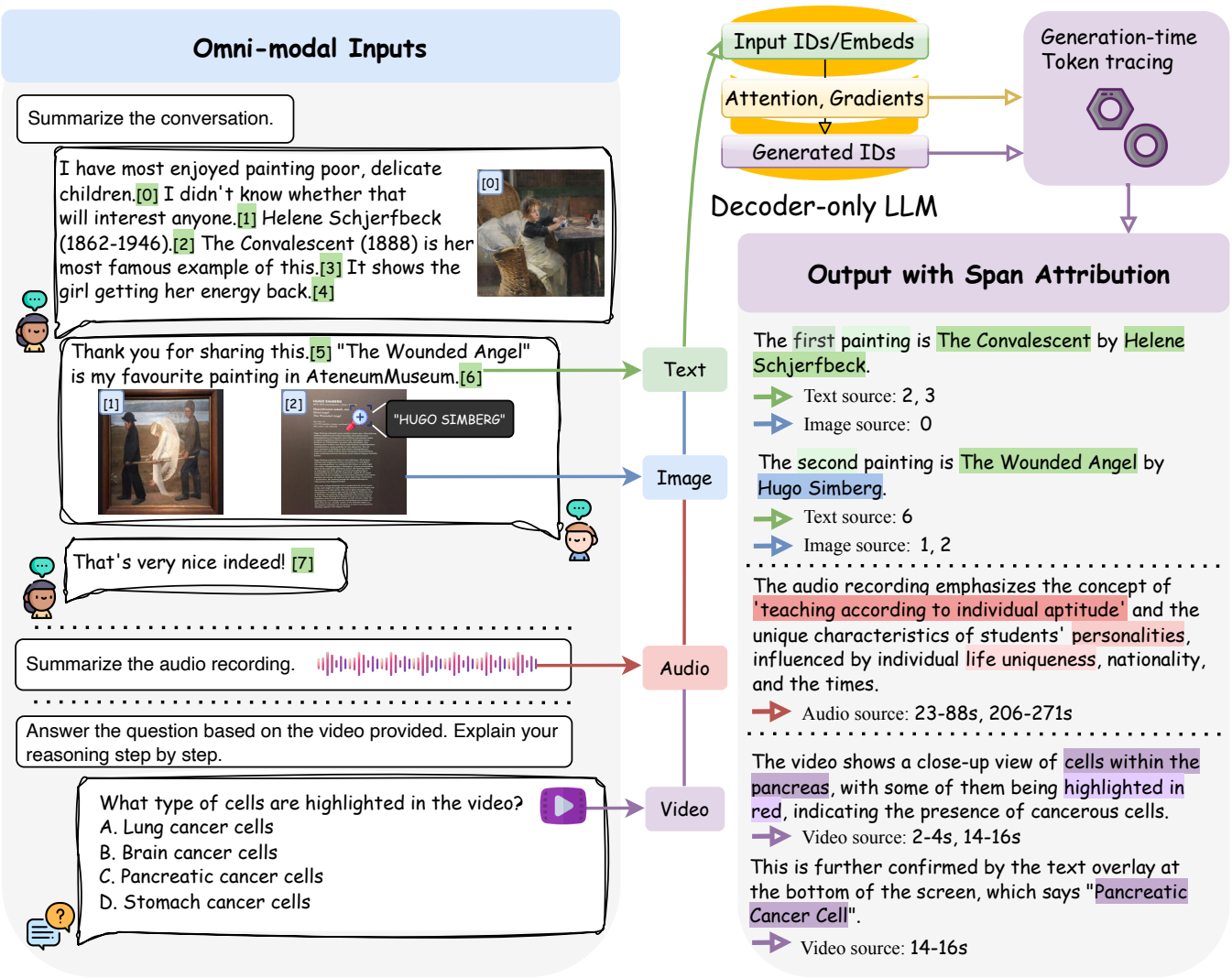

Figure 1. **OmniTrace performs generation-time attribution across modalities.** Given interleaved multimodal inputs (text, images, audio, video), an omni-modal LLM generates output tokens autoregressively. OmniTrace traces each generated token to candidate source tokens across modalities and aggregates these signals into semantically coherent output spans with concise source explanations. The framework operates online during decoding and supports heterogeneous inputs within a unified token timeline.

As a result, applying existing attribution techniques to decoder-only multimodal generation raises three challenges. First, attribution must be *generation-aware*: explanations should be defined with respect to individual decoding steps rather than a fixed prediction objective. Second, attribution must be *omni-modal*: generated tokens may depend on heterogeneous sources across modalities that share a common token timeline. Third, attribution must be *semantically interpretable*: raw token-level signals exhibit high variance across decoding steps and frequently fragment across modalities, making them unstable and semantically incoherent at the statement level.

In this work, we introduce **OmniTrace**, a lightweight and model-agnostic framework for generation-time attribution in decoder-only omni-modal LLMs. OmniTrace provides a unified protocol that converts arbitrary

token-level attribution scores into coherent source-level explanations for open-ended multimodal generation. OmniTrace is designed to be *plug-and-play*: it is orthogonal to the choice of underlying signal, including attention-based scores, gradient-derived measures, or other token alignment estimates. The framework operates directly during decoding, maps each generated token to candidate input sources across modalities, and aggregates these signals into semantically meaningful spans that correspond to human-interpretable units of the output.

We evaluate OmniTrace across decoder-only omni-modal LLMs on a diverse benchmark spanning visual, audio, and video reasoning and summarization tasks (Table 1). Our evaluation covers multi-image reasoning (Mantis-eval [18]), interleaved image–text summarization (MMDialog [19], CliConSummation [20]), audio reasoning and meeting summarization (MMAU [21], MISP [22]), and video question answering (Video-MME [23]), totaling 759 examples across heterogeneous modalities. Across all settings, generation-time span-level attribution consistently produces more stable and semantically coherent explanations than naive self-attribution and embedding-based baselines.

In summary, our contributions are:

- We formalize attribution for open-ended multimodal generation as a generation-time tracing problem over decoder-only architectures, highlighting the limitations of fixed-target and single-modality explanations.
- We introduce OmniTrace, a signal-agnostic, generation-aware framework that converts token-level attribution scores into span-level, cross-modal source explanations during decoding without retraining.
- We demonstrate that generation-aware span-level attribution produces more stable and interpretable grounding signals across multiple omni-modal LLMs and tasks.

2. Related Works

2.1. Omnimodal Large Language Models

Recent multimodal large language models [1, 2] extend decoder-only language models to support interleaved inputs spanning text, images, audio, and video. Open-sourced models such as Qwen-Omni [5, 3], MiniCPM-o [4], OmniVinci [8], OpenOmni [24] integrate modality-specific encoders (e.g., vision and speech encoders) with large language model backbones to enable unified multimodal understanding and generation. These models can perform a wide range of tasks, including multimodal question answering, summarization, and dialogue, while reasoning over heterogeneous inputs within a single conversational context.

Despite rapid progress in capability, interpretability for omnimodal generation remains largely unexplored. Existing models typically output fluent responses without explicit mechanisms to trace generated content back to supporting multimodal evidence. This lack of source attribution makes it difficult to understand how models integrate information across modalities during generation.

2.2. Attribution Methods for Transformer Models

A large body of work studies post-hoc attribution for neural networks by estimating which input features most influence a model's prediction. Most existing approaches have been developed for classification settings and encoder-based architectures, where explanations are computed with respect to a fixed objective such as a class logit or extractive span [9, 10, 11, 12, 13, 14, 15, 16, 17]. These methods typically produce token-level importance scores that quantify how strongly each input token contributes to a particular model output.

Attention-based attribution. Several approaches interpret attention weights as indicators of feature importance. For example, attention rollout and attention flow propagate attention scores across transformer layers to estimate token influence in the final prediction [9]. Subsequent work analyzes or aggregates attention patterns to explain model behavior in transformer architectures [11, 16, 17].

Gradient- and relevance-based methods. Another class of techniques derives attribution scores by propagating gradients or relevance signals from model outputs back to the input. Representative approaches include saliency maps [13], Grad-CAM [15], Gradient×Input [14], and Layer-wise Relevance Propagation (LRP) [10], which has been adapted to transformer architectures [12]. These methods quantify how sensitive the output prediction is to perturbations in each input feature, producing fine-grained token-level importance maps.

Limitations for autoregressive multimodal generation. Despite their effectiveness, existing attribution methods are primarily designed for settings where the model output is fixed and known in advance. In contrast, decoder-only large language models generate tokens sequentially, forming a growing causal graph whose structure evolves during decoding. Attribution must therefore be computed dynamically with respect to intermediate generation steps rather than a single output logit. Furthermore, modern multimodal models operate over interleaved inputs spanning text, images, audio, and video. Existing attribution techniques generally assume single-modality inputs and do not directly address how evidence should be traced across heterogeneous token types within a unified generation process.

3. OmniTrace

We now introduce OmniTrace, our generation-time attribution framework for decoder-only omni-modal language models.

3.1. Problem Formulation

We consider the problem of attributing the outputs of a decoder-only omni-modal large language model to its input sources during open-ended generation.

Inputs. Let $\mathbf{x} = (x_1, \ldots, x_n)$ denote an interleaved sequence of input tokens drawn from multiple modalities, including text, image, audio, video or other encoded representations. All modalities are embedded into a unified token space and processed jointly by the model. We assume that $\mathbf{x}$ contains identifiable source segments, such as text spans, image regions, audio or video segments, which we denote by a set of source units $\mathcal{S} = \{S_1, \ldots, S_m\}$, where each S_j corresponds to a contiguous subset of input tokens.

Generation process. Given $\mathbf{x}$, a decoder-only model generates an output sequence $\mathbf{y} = (y_1, \ldots, y_T)$ autoregressively:

$$P(\mathbf{y} \mid \mathbf{x}) = \prod_{t=1}^{T} P(y_t \mid \mathbf{x}, y_{<t}).$$

Unlike classification or extractive tasks, no fixed target or alignment is provided. Instead, attribution must be inferred from the model's internal signals at generation time.

Token-level attribution signals. For each generation step t, we assume access to a token-level attribution signal

$$a_t(i) \geq 0 \quad \text{for } i \in \{1, \ldots, n+t-1\},$$

which measures the influence of token i on the generation of y_t. Such signals may be derived from attention weights, gradients, or other model-internal statistics. We treat a_t as a generic scoring function, without assuming a particular attribution mechanism.

Generation-time attribution objective. Our goal is to map each generated token y_t to a set of source units in $\mathcal{S}$ that plausibly explain its content. However, token-level signals are often noisy and fragmented across modalities. To produce interpretable explanations, we instead define attribution at the level of *generation spans*.

Let $\mathcal{C} = \{C_1, \ldots, C_K\}$ denote a segmentation of the output sequence into semantically coherent chunks (e.g., phrases or sentences), where each C_k corresponds to a subset of generated tokens. For each chunk C_k, we seek a set of source units $\hat{\mathcal{S}}_k \subseteq \mathcal{S}$ that collectively provide sufficient evidence for that segment.

Formally, the attribution problem is to construct a mapping

$$f : C_k \mapsto \hat{\mathcal{S}}_k,$$

such that the selected sources explain the generation of the chunk while remaining concise and interpretable.

Design requirements. An effective attribution framework for omni-modal generation should satisfy four properties:

1. **Generation-aware:** attribution is defined with respect to individual decoding steps rather than a fixed output objective.
2. **Omni-modal:** attribution operates over a unified token timeline spanning multiple modalities.
3. **Span-level:** explanations are produced at semantically meaningful units of the output rather than isolated tokens.
4. **Model-agnostic:** the method should accept arbitrary token-level attribution signals enabling compatibility with diverse models and scoring mechanisms.

In the following section, we present OmniTrace, a framework that fulfills these requirements by converting token-level attribution signals into stable span-level source explanations during generation.

3.2. OmniTrace: Generation-Time Attribution Algorithm

We now present OmniTrace as a unified generation-time attribution algorithm that converts token-level signals into span-level source explanations. Algorithm 1 summarizes the full procedure. Given a prompt p and interleaved omni-modal inputs $\mathcal{M}$, the processor P converts them into a unified token sequence $\mathbf{x}$ spanning text, image, audio, and video tokens. The algorithm then traces each generated token to its most relevant source unit during autoregressive decoding and aggregates these token-level traces into semantically coherent span-level explanations.

Algorithm 1 OmniTrace: Generation-Time Span-Level Attribution for Omni-Modal LLMs

Require: Prompt p, omni-modal inputs $\mathcal{M}$ (text/image/audio/video); model M with processor P; scoring method ϕ; source curation config `cfg`
Ensure: Generated text $\mathbf{y}$ and span-level attributions $\{(C_k, \hat{\mathcal{S}}_k)\}_{k=1}^{K}$

1: $\mathbf{x} \leftarrow P(p, \mathcal{M})$ ▷ Convert prompt and inputs to token ids
2: $\mathcal{S} \leftarrow \texttt{BuildSources}(\mathbf{x}, \mathcal{M})$ ▷ Source units S_j: text spans, image regions, audio/video timestamps
3: Initialize $\mathbf{y} \leftarrow ()$ and token-to-source map $\hat{s}(\cdot)$
4: **for** $t = 1$ to T **do**
5: $\quad y_t \sim P_M(\cdot \mid \mathbf{x}, y_{<t})$ ▷ Autoregressive decoding
6: $\quad$ Obtain token-level attribution signal $a_t(i)$ using ϕ
7: $\quad \hat{s}(t) \leftarrow \arg\max_{S_j \in \mathcal{S}} \sum_{i \in S_j} a_t(i)$ ▷ Map token to most relevant source
8: $\quad c_t \leftarrow \max_{S_j \in \mathcal{S}} \sum_{i \in S_j} a_t(i)$ ▷ Confidence score
9: $\quad$ Append y_t to $\mathbf{y}$
10: **end for**
11: $\mathcal{C} \leftarrow \texttt{Chunk}(\mathbf{y})$ ▷ Segment generation into spans
12: **for all** $C_k \in \mathcal{C}$ **do**
13: $\quad$ Let T_k be token indices in C_k
14: $\quad \texttt{src_seq} \leftarrow [\hat{s}(t)]_{t \in T_k}$
15: $\quad \texttt{pos_seq} \leftarrow [\texttt{pos}_t]_{t \in T_k}$
16: $\quad \texttt{conf_seq} \leftarrow [c_t]_{t \in T_k}$
17: $\quad \hat{\mathcal{S}}_k \leftarrow \texttt{CurateSourcesWithConf}(\texttt{src_seq}, \texttt{pos_seq}, \texttt{conf_seq}, \texttt{cfg})$
18: **end for**
19: **return** $\mathbf{y}$ and $\{(C_k, \hat{\mathcal{S}}_k)\}_{k=1}^{K}$

Generation-Time Source Tracing. The first stage of OmniTrace operates directly within the decoding loop (Lines 4–10 in Algorithm 1). At each generation step t, the model produces a token y_t conditioned on $\mathbf{x}$ and the previously generated tokens $y_{<t}$. Simultaneously, we obtain a token-level attribution signal $a_t(i)$ over the causal context, where $i \in \{1, \ldots, |\mathbf{x}| + t - 1\}$. The signal $a_t(i)$ may be derived from attention weights or attention gradient-based scores, and OmniTrace remains agnostic to the specific scoring mechanism.

To obtain a source-level interpretation, we project token-level scores onto predefined source units $\mathcal{S} = \{S_1, \ldots, S_m\}$ by aggregating attribution mass within each unit:

$$\hat{s}(t) = \arg\max_{S_j \in \mathcal{S}} \sum_{i \in S_j} a_t(i).$$

Thus, each generated token y_t is traced to the source unit that receives the highest attribution mass. We additionally record a confidence score c_t which reflects the strength of evidence supporting the mapping. This generation-aware tracing ensures that attribution is computed with respect to the evolving causal context rather than a fixed target.

Span-Level Aggregation and Confidence-Based Source Selection Token-level traces are often noisy and fragmented, particularly in long-form or multimodal reasoning. To produce semantically interpretable explanations, OmniTrace aggregates token-level mappings at the level of generation spans. After decoding, the output sequence $\mathbf{y}$ is segmented into semantically coherent chunks $\mathcal{C} = \{C_1, \ldots, C_K\}$, e.g., phrases or sentences obtained via syntactic parsing.

For each span C_k, let T_k denote the indices of tokens in the span. We collect the corresponding source

Table 1. Evaluation datasets across modalities and task types.

Modality	Task Type	Dataset	# Examples	# Images
Visual	QA	Mantis-eval [18]	200	2.52
	Summarization	MMDialog [19]	157	2.73
		CliConSummation [20]	100	1.00
				Duration(s)
Audio	QA	MMAU [21]	134	12.50
	Summarization	MISP [22]	66	302.52
Video	QA	Video-MME [23]	102	32.13
Overall Total			**759**	

assignments $\{\hat{s}(t)\}_{t\in T_k}$ and confidence scores $\{c_t\}_{t\in T_k}$. This span-level pooling reduces variance in token-level attribution and stabilizes cross-modal mappings by exploiting local semantic coherence. Instead of explaining isolated tokens, OmniTrace explains complete semantic units of the generation, aligning attribution with human-interpretable statements.

Finally, OmniTrace selects a concise set of supporting sources $\hat{\mathcal{S}}_k$ for each span C_k using confidence-aware voting and temporal coherence constraints (Lines 12–18). We apply a source curation stage that filters noisy token-level signals using POS-aware weighting, confidence shaping, and run-level coherence constraints. We detail the implementation in Appendix A.1.

4. Experiments

We evaluate OmniTrace across multiple modalities and tasks to assess attribution quality, robustness, and cross-modal generalization.

4.1. Experimental Setup

We evaluate OmniTrace across visual, audio, and video tasks spanning reasoning and summarization. All experiments are conducted on decoder-only omni-modal LLMs (Qwen2.5-Omni-7B [5] and MiniCPM-o-4.5-9B [4]) on H200 GPUs under deterministic decoding to ensure reproducibility of generation-time attribution signals. Unless otherwise stated, greedy decoding is used. Details of base models are in Appendix D.1.

As show in Table 1, our evaluation covers 759 examples across heterogeneous modalities and task types.

Ground-truth attribution. Because different models produce distinct generations, ground-truth attribution must be defined relative to each model's output. Under deterministic decoding, we obtain two fixed sets of model responses (one per model). For each generated sentence, we construct a semantic attribution task: given the input sources and the generated sentence, GPT-5.2 [1] and Gemini-3 [2] are prompted to assign the supporting source units based on semantic consistency. Detailed evaluation prompt is in Appendix B.1.

Evaluation metrics. For visual-text tasks, each source unit corresponds to a discrete text span or image span. Attribution is evaluated as a multi-label prediction problem, and we report span-level F1.

For audio and video tasks, source units correspond to timestamp intervals. We discretize time into 1-second bins and compute *Time-F1*, treating each time bin as a binary label and F1 is computed in the standard way.

Human validation of LLM-as-judge. To assess the reliability of this automatic labeling procedure, human experts manually annotate 26.6% of the test set. We measure agreement between generated labels and human annotations using the same evaluation metrics. The observed alignment rate is 88.17%, indicating that the LLM-based labeling protocol provides a reliable approximation of human semantic judgments. Details can be found in Appendix B.2. Beyond attribution accuracy with respect to external labels, we evaluate whether OmniTrace reflects the model's own decision process in Appendix B.3.

4.2. Main Results

Table 2. Attribution performance across omni-modal models and tasks. $\text{OT}_{\text{AttMean}}$, $\text{OT}_{\text{RawAtt}}$, and $\text{OT}_{\text{AttGrads}}$ denote OmniTrace instantiated with mean-pooled attention, raw attention, and gradient-based scoring signals, respectively. Post-hoc heuristics include self-attribution, embedding-based similarity (processor embeddings and CLIP), and random assignment. † indicates results not reported due to computational constraints (gradient-based attribution is memory-intensive for long audio/video inputs). × indicates that a method is not applicable to continuous timestamp attribution in audio/video tasks. Detailed implementation and discussions the baselines are provided in Appendix C.2.

Method	Visual Tasks			Audio Tasks		Video Tasks
	Summ.		QA	Summ.	QA	QA
	Text F1	Image F1	Image F1	Time F1	Time F1	Time F1
Qwen2.5-Omni-7B generation-time attribution						
$\text{OT}_{\text{AttMean}}$	**75.66**	**76.59**	56.60	**83.12**	**49.90**	**40.16**
$\text{OT}_{\text{RawAtt}}$	72.51	51.82	**65.44**	76.69	47.64	36.53
$\text{OT}_{\text{AttGrads}}$	67.70	42.24	65.02	†	47.56	†
Post-hoc heuristics						
Self-Attribution	9.25	40.60	61.03	4.43	29.01	13.67
$\text{Embed}_{\text{processor}}$	17.30	14.55	36.88	×	×	×
$\text{Embed}_{\text{CLIP}}$	17.20	3.54	6.32	×	×	×
Random	10.98	8.38	24.70	×	×	×
MiniCPM-o 4.5-9B generation-time attribution						
$\text{OT}_{\text{AttMean}}$	30.57	75.43	37.00	33.52	**46.94**	**22.85**
$\text{OT}_{\text{RawAtt}}$	**37.32**	**76.46**	**45.41**	**49.21**	41.06	21.59
Post-hoc heuristics						
Self-Attribution	9.06	66.53	39.39	0.08	34.66	18.26
$\text{Embed}_{\text{processor}}$	18.02	7.14	5.98	×	×	×
$\text{Embed}_{\text{CLIP}}$	17.98	5.55	5.32	×	×	×
Random	12.05	10.03	22.96	×	×	×

Table 2 reports attribution performance across modalities, models, and scoring methods. We report performance of OmniTrace with three orthogonal underlying scoring functions: attention-based scores including AttMean, RawAtt [9] and attention gradients-based scorer AttGrads [14]. We dicuss the choice of our scoring functions in Appendix C.1. We also report performance from four other baselines, including model self-attribution, embedding-based heuristics and a random baseline. Implementation details and discussion are in Appendix C.2.

Table 3. Ablation study of the source curation framework. We report Precision, Recall, and F1 for text, image, audio, and video attribution.

Method	Text			Image			Audio			Video		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1
Full Model (Default)	83.06	75.10	75.66	84.19	75.12	76.59	65.96	47.27	49.90	56.57	47.08	40.16
w/o POS Weighting	82.80	77.13	76.69	23.70	19.94	20.79	74.66	43.94	50.07	56.85	41.30	37.46
w/o Confidence Weight	76.82	78.82	74.59	22.88	18.85	19.88	70.95	46.26	50.83	56.36	43.50	38.82
w/o Confidence	68.13	80.96	70.85	22.84	18.94	19.91	74.26	42.33	48.69	56.48	38.21	35.80
w/o Run Coherence	83.09	75.52	75.93	22.88	18.85	19.88	74.26	42.23	48.69	56.48	38.20	35.79
w/o $p_{\min}$ Filtering	82.92	75.70	76.00	22.88	18.85	19.88	74.21	42.49	48.85	56.62	38.77	36.22

Overall performance. Across both Qwen2.5-Omni and MiniCPM-o-4.5, all OmniTrace variants substantially outperform post-hoc baselines. On Qwen, $\text{OT}_{\text{Attmean}}$ performs best on all tasks except for visual QA. Meanwhile, for MiniCPM, $\text{OT}_{\text{RawAtt}}$ consistently yields the strongest results for visual and audio tasks. These trends suggest that while the choice of token-level scoring (attention pooling vs. raw attention vs. gradient-based signals) affects absolute performance, the core improvements stem from the generation-aware tracing and span-level aggregation framework, which remains robust across scoring instantiations.

Cross-modal generalization. OmniTrace generalizes across heterogeneous source representations. For audio and video tasks, attribution operates over continuous timestamp intervals rather than discrete span labels. Despite this shift in label space, generation-time attribution achieves strong Time-F1 scores, e.g., 49.90 (Qwen audio QA) and 46.94 (MiniCPM audio QA), while post-hoc self-attribution remains substantially lower. This demonstrates that the framework is not tied to discrete text/image units, but extends naturally to temporal grounding in continuous domains.

4.2.1. Effect of Confidence-Based Filtering

We conduct furthur ablation study using the strongest setting: Qwen2.5-Omni-7B with `attmean` as the scoring function, ablating each component of the source curation framework independently.

Overall impact. As shown in Table 3, removing any filtering component leads to consistent degradation, with a particularly dramatic effect on image attribution. While text and time F1 varies within a relatively narrow range, image F1 drops sharply from 76.59 (full model) to around 20 under nearly all ablations. This collapse indicates that raw token-level tracing alone is insufficient for reliable cross-modal grounding. In contrast, the proposed confidence-aware filtering pipeline is essential for suppressing spurious visual assignments and concentrating attribution mass on truly supported evidence. Detailed analysis on the effect of each ablated parameter can be found in Appendix A.2.

Interpretation. Across all ablations, image attribution is markedly more sensitive to filtering mechanisms than text attribution. This asymmetry reflects the denser and more redundant nature of textual evidence, in contrast to the sparse and fragile signals characteristic of visual grounding. Collectively, these results demonstrate that the proposed curation framework: integrating POS-aware semantic weighting, confidence shaping, run-level coherence, and minimum-mass filtering, is indispensable for transforming noisy token-level traces into stable, compact, and interpretable span-level cross-modal explanations.

4.2.2. *Effect of ASR Segmentation*

Audio attribution requires mapping generated spans to temporally localized speech segments. We investigate how automatic speech recognition (ASR) quality and segmentation granularity affect attribution performance.

We compare three ASR systems of varying quality: Paraformer [25], Scribe v2, and Whisper [26] against a raw token baseline without ASR segmentation. Details of ASR models used is in Appendix D.2. For ASR-based methods, speech is segmented into semantically coherent chunks with timestamps, which serve as candidate source units for attribution. In the raw-token setting, the audio input is processed without semantic segmentation, resulting in fine-grained and temporally fragmented token sequences.

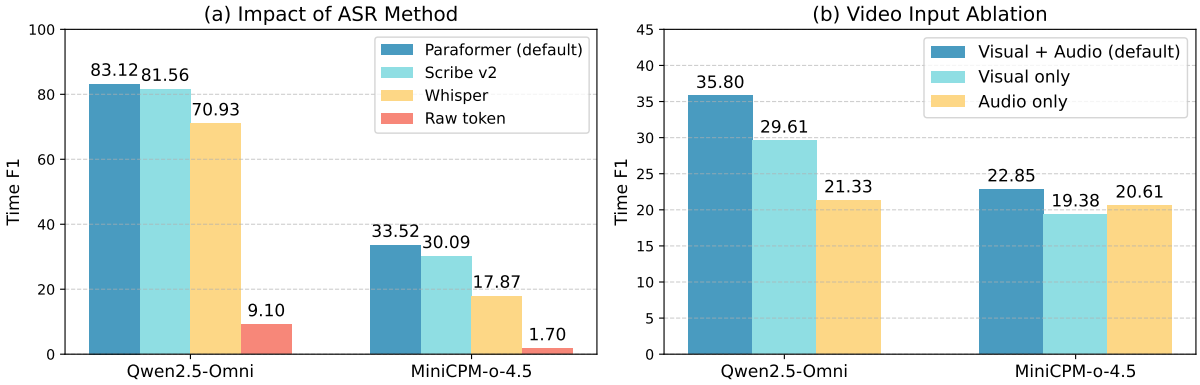

Figure 2. **Effect of input modality segmentation and availability on attribution performance.** (a) Impact of ASR segmentation quality on audio attribution. Time-F1 on audio summarization using different ASR systems. High-quality ASR segmentation (Paraformer, Scribe v2) substantially improves attribution accuracy, while raw token inputs without semantic segmentation lead to severe degradation. (b) Video input ablation. We compare attribution performance when using both visual and audio signals (default) versus using only one modality. Using both modalities consistently yields the best performance, indicating that video attribution benefits from jointly leveraging complementary visual and auditory evidence.

Figure 2(a) reports Time-F1 on the audio summarization task for Qwen2.5-Omni and MiniCPM-o-4.5. Several observations emerge.

ASR segmentation is critical. Using high-quality ASR dramatically improves attribution performance. Paraformer achieves 83.12 Time-F1 on Qwen and 33.52 on MiniCPM, while Scribe v2 yields comparable performance. In contrast, removing semantic segmentation (raw tokens) causes severe degradation. This confirms that coherent temporal segmentation is essential for stable span-level attribution.

Model robustness differences. Qwen consistently outperforms MiniCPM across all ASR conditions, but both models exhibit the same trend: attribution performance is highly sensitive to speech segmentation quality. This indicates that the dependency on structured temporal units is not specific to a particular model architecture.

4.2.3. *Effect of Visual and Audio Source*

Video inputs naturally contain both visual frames and audio tracks. In OmniTrace, attribution is computed jointly over both modalities, and the final source set is obtained by taking the union of curated visual and

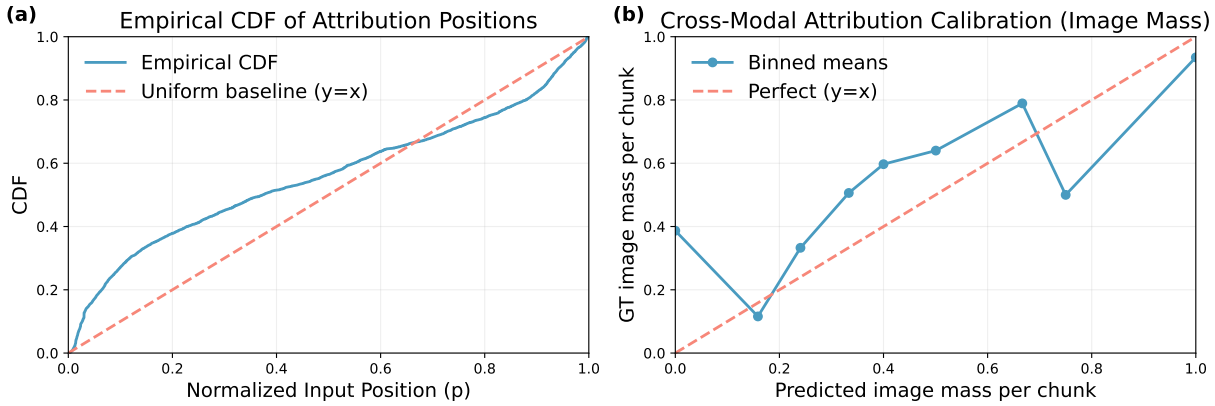

Figure 3. **Positional and cross-modal attribution behavior.** (a) Empirical CDF of normalized attribution positions. The curve above the diagonal indicates early-position bias. (b) Cross-modal calibration between predicted and ground-truth image mass. Deviations from the diagonal reflect regime-dependent calibration effects rather than a strong global modality bias.

audio sources. To evaluate the contribution of each modality, we perform an ablation where attribution is computed using only visual inputs or only audio inputs.

Figure 2(b) reports Time-F1 on the video QA task under three settings: (i) visual + audio (default), (ii) visual only, and (iii) audio only.

Complementary modalities improve attribution. Using both visual and audio inputs consistently yields the best attribution performance. For Qwen2.5-Omni, the full multimodal setting achieves 35.80 Time-F1, compared to 29.61 using only visual inputs and 21.33 using only audio inputs. This demonstrates that both modalities provide complementary evidence for locating supporting sources.

Different models rely on modalities differently. For MiniCPM-o-4.5, visual-only attribution performs slightly worse than the multimodal setting, while audio-only attribution remains competitive with visual-only. This suggests that different MLLMs may rely on modality signals differently, but combining modalities still produces the most reliable attribution overall.

These results highlight the importance of preserving multimodal context during attribution, as restricting the input to a single modality can remove critical supporting evidence.

5. Analysis

We further analyze the behavior of OmniTrace to understand potential attribution biases and its relationship with generation quality.

5.1. Positional and Modality Attribution Bias

We analyze whether attribution produced by OmniTrace exhibits systematic biases with respect to input position or modality. Figure 3 summarizes the results.

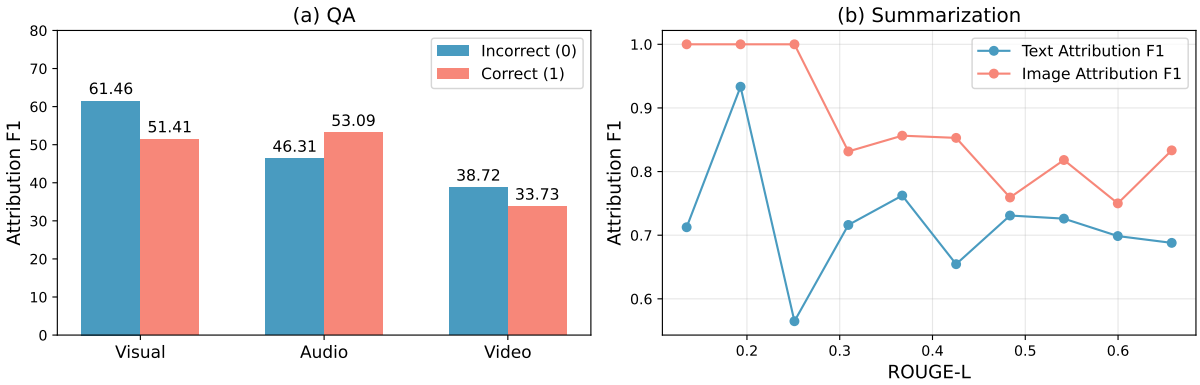

Figure 4. **Relationship between generation quality and attribution quality.** (a) Attribution F1 for QA tasks grouped by whether the generated answer is correct or incorrect in QA tasks. (b) Attribution performance on visual summarization as a function of generation quality measured by ROUGE-L. While attribution quality generally improves with higher generation quality in some modalities (e.g., audio), the relationship is not strictly monotonic, indicating that attribution reflects grounding behavior rather than merely the correctness of the final output.

Positional attribution bias. To examine whether attribution is uniformly distributed across the input sequence in summarization tasks, we analyze the normalized position $p \in [0, 1]$ of selected source units.

Figure 3(a) shows the empirical CDF of attribution positions. If attribution were position-neutral, the curve would follow the uniform diagonal. Instead, the empirical CDF lies consistently above the baseline, indicating a noticeable early-token bias. The mean normalized attribution position is $\bar{p} = 0.44 < 0.5$, confirming that attribution mass concentrates disproportionately in earlier portions of the input sequence.

Notably, this bias appears even in summarization tasks, where supporting evidence may originate from any part of the input. This suggests that decoder-only omni-modal models may preferentially ground generation in earlier context segments, potentially reflecting attention dynamics or positional priors in autoregressive decoding.

Cross-modal attribution bias. We next analyze whether attribution systematically favors one modality when both visual and textual evidence are available.

Figure 3(b) shows calibration between predicted and ground-truth image attribution mass. If attribution were modality-neutral, the curve would follow the diagonal. Instead, we observe a non-linear calibration pattern. In the moderate regime (predicted image mass between 0.3 and 0.6), the empirical curve lies slightly above the diagonal, indicating mild under-attribution to image evidence in mixed-modality chunks. Conversely, at high predicted image mass (above 0.7), the curve falls below the diagonal, suggesting occasional overconfidence when assigning image-dominant explanations.

5.2. Generation Quality vs Attribution Quality

We analyze whether attribution quality correlates with generation quality across tasks. For QA tasks, generation quality is measured by answer correctness, while for summarization we use ROUGE-L as a continuous measure of generation quality.

Figure 4(a) shows attribution F1 for samples where the generated answer is correct versus incorrect. Interestingly, attribution quality does not always increase when answers are correct. For example, in visual and

video QA, attribution scores remain comparable or even slightly higher for incorrect predictions, suggesting that the model may still attend to relevant evidence even when the final answer is wrong. In contrast, audio QA shows improved attribution quality for correct responses, indicating that accurate grounding in temporal segments may contribute more directly to answer correctness.

Figure 4(b) examines summarization tasks using ROUGE-L as a proxy for generation quality. We observe a weak positive correlation between ROUGE-L and attribution F1, particularly for text attribution. However, the relationship remains noisy and non-monotonic, indicating that attribution quality is not simply a byproduct of better generation quality.

Overall, these results suggest that attribution captures aspects of the model's evidence-grounding behavior that are partially independent of the final output quality. This highlights the importance of evaluating attribution explicitly rather than assuming it improves automatically with generation accuracy.

6. Conclusion

We introduced **OmniTrace**, a generation-time attribution framework for decoder-only omni-modal large language models. Unlike traditional attribution methods designed for fixed outputs, OmniTrace traces token-level influence during autoregressive decoding and aggregates these signals into coherent span-level source explanations across text, image, audio, and video inputs. The framework is model-agnostic and supports multiple scoring signals, including attention-based and gradient-based attribution.

Experiments across visual, audio, and video tasks demonstrate that OmniTrace consistently improves token-to-source alignment compared with post-hoc heuristics. Our analysis further shows that attribution quality depends on meaningful source segmentation and multimodal context, while remaining partially independent from generation correctness. Additional studies reveal positional grounding tendencies and calibration behavior across modalities, providing insights into how omni-modal LLMs utilize input evidence during generation.

Overall, our results suggest that generation-time source tracing offers a practical and scalable approach for interpreting open-ended multimodal generation. We hope OmniTrace provides a useful foundation for improving transparency, debugging, and trustworthiness in future omni-modal language models.

References

[1] Aaditya Singh, Adam Fry, Adam Perelman, Adam Tart, Adi Ganesh, Ahmed El-Kishky, Aidan McLaughlin, Aiden Low, AJ Ostrow, Akhila Ananthram, et al. OpenAI GPT-5 System Card. *arXiv preprint arXiv:2601.03267*, 2026.

[2] Google Gemini Team. A new era of intelligence with gemini 3. https://blog.google/products-and-platforms/products/gemini/gemini-3, 2025. Google AI Blog, Accessed Jan 16, 2026.

[3] Jin Xu, Zhifang Guo, Hangrui Hu, Yunfei Chu, Xiong Wang, Jinzheng He, Yuxuan Wang, Xianzhong Shi, Ting He, Xinfa Zhu, Yuanjun Lv, Yongqi Wang, Dake Guo, He Wang, Linhan Ma, Pei Zhang, Xinyu Zhang, Hongkun Hao, Zishan Guo, Baosong Yang, Bin Zhang, Ziyang Ma, Xipin Wei, Shuai Bai, Ke Chen, Xue Lian Liu, Peng Wang, Ming Yang, Dayiheng Liu, Xingzhang Ren, Bo Zheng, Rui Men, Fan Zhou, Bowen Yu, Jianxin Yang, Le Yu, Jing-Jun Zhou, and Junyang Lin. Qwen3-omni technical report. *ArXiv*, abs/2509.17765, 2025. URL https://api.semanticscholar.org/CorpusID:281420796.

[4] Tianyu Yu, Zefan Wang, Chongyi Wang, Fuwei Huang, Wenshuo Ma, Zhihui He, Tianchi Cai, Weize Chen, Yuxiang Huang, Yuanqian Zhao, et al. Minicpm-v 4.5: Cooking efficient mllms via architecture, data, and training recipe. *arXiv preprint arXiv:2509.18154*, 2025.

[5] Jin Xu, Zhifang Guo, Jinzheng He, Hangrui Hu, Ting He, Shuai Bai, Keqin Chen, Jialin Wang, Yang Fan, Kai Dang, et al. Qwen2. 5-omni technical report. *arXiv preprint arXiv:2503.20215*, 2025.

[6] Wenwen Tong, Hewei Guo, Dongchuan Ran, Jiangnan Chen, Jiefan Lu, Kaibin Wang, Keqiang Li, Xiaoxu Zhu, Jiakui Li, Kehan Li, et al. Interactiveomni: A unified omni-modal model for audio-visual multi-turn dialogue. *arXiv preprint arXiv:2510.13747*, 2025.

[7] Abdelrahman Abouelenin, Atabak Ashfaq, Adam Atkinson, Hany Awadalla, Nguyen Bach, Jianmin Bao, Alon Benhaim, Martin Cai, Vishrav Chaudhary, Congcong Chen, et al. Phi-4-mini technical report: Compact yet powerful multimodal language models via mixture-of-loras. *arXiv preprint arXiv:2503.01743*, 2025.

[8] Hanrong Ye, Chao-Han Huck Yang, Arushi Goel, Wei Huang, Zhen Wan, Jinchuan Tian, An-Chieh Cheng, Ligeng Zhu, Yuanhang Su, Yuming Lou, Yong-Xiang Lin, Dong Yang, Sreyan Ghosh, Zhijian Liu, Yukang Chen, Ehsan Jahangiri, Ambrish Dantrey, Daguang Xu, Ehsan Hosseini-Asl, Seyed Danial Mohseni Taheri, Vidya Nariyambut Murali, Sifei Liu, Yao Lu, Oluwatobi Olabiyi, Yu-Chiang Frank Wang, Rafael Valle, Bryan Catanzaro, Andrew Tao, Song Han, Jan Kautz, Hongxu Yin, and Pavlo Molchanov. Omnivinci: Enhancing architecture and data for omni-modal understanding LLM. In *The Fourteenth International Conference on Learning Representations*, 2026. URL https://openreview.net/forum?id=DZeic3NpHy.

[9] Samira Abnar and Willem Zuidema. Quantifying attention flow in transformers. In Dan Jurafsky, Joyce Chai, Natalie Schluter, and Joel Tetreault, editors, *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4190–4197, Online, July 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.acl-main.385. URL https://aclanthology.org/2020.acl-main.385/.

[10] Sebastian Bach, Alexander Binder, Grégoire Montavon, Frederick Klauschen, Klaus-Robert Müller, and Wojciech Samek. On pixel-wise explanations for non-linear classifier decisions by layer-wise relevance propagation. *PLoS ONE*, 10(7):e0130140, July 2015. doi: 10.1371/journal.pone.0130140.

[11] Elena Voita, David Talbot, Fedor Moiseev, Rico Sennrich, and Ivan Titov. Analyzing multi-head self-attention: Specialized heads do the heavy lifting, the rest can be pruned. In Anna Korhonen, David Traum, and Lluís Màrquez, editors, *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 5797–5808, Florence, Italy, July 2019. Association for Computational Linguistics. doi: 10.18653/v1/P19-1580. URL https://aclanthology.org/P19-1580/.

[12] Hila Chefer, Shir Gur, and Lior Wolf. Generic attention-model explainability for interpreting bi-modal and encoder-decoder transformers. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 397–406, 2021.

[13] Bolei Zhou, Aditya Khosla, Agata Lapedriza, Aude Oliva, and Antonio Torralba. Learning deep features for discriminative localization. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2921–2929, 2016.

[14] Oren Barkan, Edan Hauon, Avi Caciularu, Ori Katz, Itzik Malkiel, Omri Armstrong, and Noam Koenigstein. Grad-sam: Explaining transformers via gradient self-attention maps. In *Proceedings*

of the 30th ACM International Conference on Information & Knowledge Management, CIKM '21, page 2882–2887, New York, NY, USA, 2021. Association for Computing Machinery. ISBN 9781450384469. doi: 10.1145/3459637.3482126. URL https://doi.org/10.1145/3459637.3482126.

[15] Ramprasaath R Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. Grad-cam: visual explanations from deep networks via gradient-based localization. *International journal of computer vision*, 128(2):336–359, 2020.

[16] Yao Qiang, Deng Pan, Chengyin Li, Xin Li, Rhongho Jang, and Dongxiao Zhu. Attcat: explaining transformers via attentive class activation tokens. In *Proceedings of the 36th International Conference on Neural Information Processing Systems*, NIPS '22, Red Hook, NY, USA, 2022. Curran Associates Inc. ISBN 9781713871088.

[17] Linxin Song, Yan Cui, Ao Luo, Freddy Lecue, and Irene Li. Better explain transformers by illuminating important information. In Yvette Graham and Matthew Purver, editors, *Findings of the Association for Computational Linguistics: EACL 2024*, pages 2048–2062, St. Julian's, Malta, March 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-eacl.138. URL https://aclanthology.org/2024.findings-eacl.138/.

[18] Dongfu Jiang, Xuan He, Huaye Zeng, Cong Wei, Max W.F. Ku, Qian Liu, and Wenhu Chen. Mantis: Interleaved multi-image instruction tuning. *Transactions on Machine Learning Research*, 2024, 2024. URL https://openreview.net/forum?id=skLtdUVaJa.

[19] Jiazhan Feng, Qingfeng Sun, Can Xu, Pu Zhao, Yaming Yang, Chongyang Tao, Dongyan Zhao, and Qingwei Lin. Mmdialog: A large-scale multi-turn dialogue dataset towards multi-modal open-domain conversation. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7348–7363, 2023.

[20] Abhisek Tiwari, Anisha Saha, Sriparna Saha, Pushpak Bhattacharyya, and Minakshi Dhar. Experience and evidence are the eyes of an excellent summarizer! towards knowledge infused multi-modal clinical conversation summarization. In *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management*, CIKM '23, page 2452–2461, New York, NY, USA, 2023. Association for Computing Machinery. ISBN 9798400701245. doi: 10.1145/3583780.3614870. URL https://doi.org/10.1145/3583780.3614870.

[21] S Sakshi, Utkarsh Tyagi, Sonal Kumar, Ashish Seth, Ramaneswaran Selvakumar, Oriol Nieto, Ramani Duraiswami, Sreyan Ghosh, and Dinesh Manocha. MMAU: A massive multi-task audio understanding and reasoning benchmark. In *The Thirteenth International Conference on Learning Representations*, 2025. URL https://openreview.net/forum?id=TeVAZXr3yv.

[22] Ming Gao, Shilong Wu, Hang Chen, Jun Du, Chin-Hui Lee, Shinji Watanabe, Jingdong Chen, Siniscalchi Sabato Marco, and Odette Scharenborg. The multimodal information based speech processing (misp) 2025 challenge: Audio-visual diarization and recognition. *arXiv preprint arXiv:2505.13971*, 2025.

[23] Chaoyou Fu, Yuhan Dai, Yongdong Luo, Lei Li, Shuhuai Ren, Renrui Zhang, Zihan Wang, Chenyu Zhou, Yunhang Shen, Mengdan Zhang, et al. Video-mme: The first-ever comprehensive evaluation benchmark of multi-modal llms in video analysis. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 24108–24118, 2025.

[24] Run Luo, Ting-En Lin, Haonan Zhang, Yuchuan Wu, Xiong Liu, Yongbin Li, Longze Chen, Jiaming Li, Lei Zhang, Xiaobo Xia, Hamid Alinejad-Rokny, Fei Huang, and Min Yang. Openomni: Advancing open-source omnimodal large language models with progressive multimodal alignment and real-time emotional speech synthesis. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025. URL https://openreview.net/forum?id=4iehXI36QG.

[25] Zhifu Gao, Shiliang Zhang, Ian McLoughlin, and Zhijie Yan. Paraformer: Fast and accurate parallel transformer for non-autoregressive end-to-end speech recognition. *arXiv preprint arXiv:2206.08317*, 2022.

[26] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision. In *International conference on machine learning*, pages 28492–28518. PMLR, 2023.

A. Source curation

A.1. Detailed implementation

For tokens within a span, we compute weighted votes that combine: (i) part-of-speech–dependent semantic weights, (ii) confidence scores c_t (optionally exponentiated), and (iii) run-length–based coherence signals that favor temporally contiguous source assignments.

Formally, the curation function

$$\hat{\mathcal{S}}_k = \texttt{CurateSourcesWithConf}\big(\{\hat{s}(t)\}_{t\in T_k}, \{\texttt{pos}_t\}_{t\in T_k}, \{c_t\}_{t\in T_k}, \texttt{cfg}\big)$$

selects the minimal set of source units whose normalized attribution mass satisfies coverage and stability constraints. This procedure filters spurious token-level fluctuations, enforces cross-token consistency, and yields concise span-level explanations. .

Together, these three stages transform arbitrary token-level attribution signals into stable, generation-aware span-level source explanations, satisfying the design requirements outlined in Section 3.1.

```python
def curate_sources_with_conf(
    source_ids: List[int],
    pos: List[str],
    conf: List[float],
    cfg: SourceCurationConfig = DEFAULT_CURATE_CFG,
) -> List[int]:
    T = len(source_ids)
    if T == 0:
        return []
    if not (T == len(pos) == len(conf)):
        raise ValueError("source_ids, pos, conf must have the same length.")

    vote: List[float] = []
    for p, c in zip(pos, conf):
        pw = POS_W.get(p, 0.3)
        cw = (max(c, 0.0) ** cfg.gamma)
        vote.append(pw * cw)

    total = float(sum(vote))
    if total <= 0:
        return []

    mass = defaultdict(float)
    for s, v in zip(source_ids, vote):
        mass[s] += v
    p_mass = {s: m / total for s, m in mass.items()}

    run_max = defaultdict(float)
    cur_s, cur_run = source_ids[0], vote[0]
    for i in range(1, T):
        if source_ids[i] == cur_s:
            cur_run += vote[i]
        else:
            run_max[cur_s] = max(run_max[cur_s], cur_run)
            cur_s, cur_run = source_ids[i], vote[i]
    run_max[cur_s] = max(run_max[cur_s], cur_run)
    run_frac = {s: run_max[s] / total for s in mass.keys()}

    def score(s: int) -> float:
        return cfg.alpha * p_mass[s] + (1.0 - cfg.alpha) * run_frac[s]

    ranked = sorted(p_mass.keys(), key=score, reverse=True)

    selected: List[int] = []
```

```
    cum = 0.0
    for s in ranked:
        strong_run = run_frac[s] >= cfg.run_min
        if p_mass[s] < cfg.p_min and not strong_run:
            continue
        selected.append(s)
        cum += p_mass[s]
        if cum >= cfg.coverage:
            break

    return selected
```

A.2. Ablation study

Below, we provide a detailed analysis of each ablation setting, clarifying the functional role of each component in the source curation pipeline and its empirical impact.

Role of POS-aware weighting. POS-aware weighting assigns higher contribution to semantically informative tokens (e.g., nouns, proper nouns, numerals) and downweights function words and punctuation during attribution mass aggregation. This mechanism ensures that source selection is driven primarily by content-bearing tokens rather than syntactic scaffolding. Disabling POS-aware weighting produces only minor changes in text, audio and video attribution. However, image F1 decreases drastically to 20.79. This indicates that visual grounding is particularly sensitive to semantic token emphasis: without syntactic weighting, low-information tokens contribute equally, causing attribution mass to disperse across irrelevant image sources and destabilizing cross-modal assignments.

Role of confidence shaping. Confidence shaping modulates token contributions according to the strength of their token-to-source linkage, amplifying high-certainty alignments while attenuating weak or ambiguous ones. Concretely, the exponent parameter γ sharpens attribution mass toward confident mappings. Without confidence modulation, attribution mass becomes diffuse across competing candidates, leading to over-selection of spurious image sources. The larger degradation in image F1 suggests that visual grounding signals are inherently sparser and noisier than textual alignments, making confidence-aware filtering critical in cross-modal scenarios.

Role of run-level coherence. Run-level coherence promotes temporal consistency by favoring source chunks that receive sustained support across consecutive generated tokens. This mechanism captures the intuition that semantically grounded evidence often corresponds to contiguous spans (e.g., multiple tokens describing the same object). Without enforcing temporal consistency, scattered token-level assignments accumulate across unrelated image sources, undermining stable visual grounding. These results indicate that coherence constraints are particularly important for consolidating cross-token visual evidence.

Role of threshold filtering ($p_{\min}$). The minimum-mass threshold filters out source chunks whose aggregated attribution mass falls below a predefined proportion, preventing low-impact assignments from entering the final selection. This confirms that weak token-level signals—common in visual grounding— must be explicitly filtered to avoid over-selection of irrelevant image spans. Threshold filtering therefore acts as a crucial denoising step in the multimodal setting.

B. Evaluation

B.1. LLM-as-judge prompts

B.1.1. Visual Tasks

We used GPT5.2 [1] for visual tasks labeling. We give the LLM judge the chunked source as well as chunked model generation and ask it to treat it as a multi-label prediction problem.

```
prompt = '''You are an annotation assistant for SOURCE ATTRIBUTION.

You will be given:
1) An image that contains the source conversation with image IDs shown on the images.
2) A model-generated summary (split into sentences, each sentence has a sentence index).
3) The source text chunks listed below, each with a numeric text ID.

Task:
For EACH generated summary sentence, decide which source evidence supports it.
- "text_source": a list of TEXT IDs from the provided text chunks that directly support
    that sentence.
- "image_source": a list of IMAGE IDs visible in the provided image that directly support
    that sentence.
Return ONLY IDs that are explicitly supported by the source. Do NOT invent IDs.

Attribution rules:
- Include an ID only if it is necessary to justify the sentence content.
- If the sentence is supported by multiple text or image chunks, include all relevant IDs.
- If the sentence refers to visual content (objects, actions, clothing, scene, etc.),
    include the relevant image IDs.
- If a sentence is NOT directly supported by any source evidence, output an empty list for
     that field.
- Do NOT use the instruction chunk (e.g., "Summarize the conversation") as evidence unless
     the sentence is literally about the instruction.
- Do NOT merge evidence across sentences: each sentence gets its own dict.
- IDs must be integers (no quotes), in ascending order, with no duplicates.

Output format (STRICT):
Return a JSON array with length == number of generated summary sentences.
Each element corresponds to the sentence index order (0..N-1) and must be exactly:
{{"image_source": [...], "text_source": [...]}}

Do not output any extra keys, explanations, markdown, or trailing text.

Model generated response (sentences indexed):
{model_generation}

Text source chunks (format: [text_id] text):
{text_source_chunks}
'''
```

B.1.2. Audio and Video Tasks

Since Audio and Video tasks has continuous labels in timeline, we are not able to provide chunked source as in the visual-text setting. We give the judge the raw audio/video source and ask it output relevant timespans for each generated chunk.

```
prompt = '''You are an annotation assistant for SOURCE ATTRIBUTION.

You will be given:
1) One AUDIO/VIDEO file that contains the source content.
2) A model-generated summary (split into sentences, each sentence has a sentence index).
```

```
Task:
For EACH generated summary sentence, decide which source evidence supports it.
- "audio/video_source": a list of timestamp spans [start_sec, end_sec] in the AUDIO/VIDEO
    that directly support that sentence.

Timestamp labeling requirements:
- Timestamps are continuous spans in SECONDS.
- Use a list of spans to allow multiple supporting regions, e.g. [[0,2], [6,8]].
- Each span must satisfy: start_sec < end_sec.
- Use minimal coverage: include ONLY the smallest time ranges that contain the relevant
    evidence.
  - Do NOT label the entire audio (e.g., [0,100]) unless the sentence truly requires the
      entire audio.
- Prefer a small number of short spans rather than one very long span.
- Round timestamps to 1s precision and keep them consistent.

Attribution rules:
- Include a timestamp span only if it is necessary to justify the sentence content.
- If the sentence is supported by multiple audio/video regions, include all relevant
    evidence.
- If a sentence is NOT directly supported by any source evidence in a modality, output an
    empty list for that field.
- Do NOT use the instruction chunk (e.g., "Summarize the conversation") as evidence unless
     the sentence is literally about the instruction.
- Do NOT merge evidence across sentences: each sentence gets its own dict.
- audio/video_source spans must be sorted by start time, non-overlapping when possible (
    merge overlaps if they touch or overlap).

Output format (STRICT):
Return a JSON array with length == number of generated summary sentences.
Each element corresponds to the sentence index order (0..N-1) and must be exactly:
{{"audio/video_source": [[start,end], ...]}}

Do not output any extra keys, explanations, markdown, or trailing text.

Model generated response (sentences indexed):
{model_generation}

STRICT OUTPUT CONTRACT:
- Output MUST be valid JSON only (no markdown, no code fences).
- Do not include ```json or ``` anywhere.
- Output must start with '[' and end with ']'.
- No extra text before or after the JSON.
- Keep it compact on one line.
'''
```

B.2. Human study

To verify the quality of LLM judge labels, we hired four human experts to manually go through around 26.6% of the data using Qwen2.5-Omni responses, and report the agreement using the same metric: span-level F1 for visual and time F1 for audio and video. Results are reported in Table 4.

B.3. Faithfulness Verification

Beyond attribution accuracy with respect to external labels, we evaluate whether OmniTrace reflects the model's own decision process. Specifically, for multiple-choice QA tasks, we test whether the attribution assigned to the model's final answer is consistent with the option it selects.

Table 4. Agreement between human annotations and LLM-judge labels on a subset of the evaluation data (26.6%). Metrics follow the same definitions used in the main evaluation: span-level F1 for visual attribution and time-based F1 for audio/video attribution.

Agreement	**Visual**			**Audio**	**Video**
Task	Summarization		QA	QA	QA
Metric	Text F1	Image F1	Image F1	Time F1	Time F1
Score	83.04	94.30	93.80	81.75	85.28
# Samples	61	61	40	40	40

Option-consistency evaluation. All QA datasets are formulated as multiple-choice questions, where each option corresponds to a discrete source text chunk. For each sample, we extract the model's predicted option $\hat{o}$ from the generated response using deterministic parsing. Importantly, this evaluation does not depend on whether the prediction is correct; it tests whether attribution reflects the model's chosen answer, not whether the answer matches ground truth.

Let C_k denote the generation span containing the final answer statement. For each option o, let S_o denote the source chunk corresponding to that option. Using the token-level mappings and confidence-weighted votes defined in Section 3.1, we compute the option-restricted attribution mass:

$$\text{Mass}(S_o) = \sum_{t \in T_k} w_t \cdot \mathbf{1}[\hat{s}(t) = S_o],$$

where T_k are tokens in the answer span, $\hat{s}(t)$ is the source traced for token t, and w_t is the confidence- and POS-weighted vote. We then define the predicted attribution option as

$$\tilde{o} = \arg\max_{o} \text{Mass}(S_o).$$

Top-1 Option Consistency. We report *Top-1 consistency*, defined as

$$\text{Consistency} = \mathbb{1}[\tilde{o} = \hat{o}],$$

averaged over samples. This metric measures whether the option receiving the highest attribution mass matches the option explicitly selected by the model.

Results. On the 200 visual QA samples, OmniTrace (Qwen + $\text{OT}_{\text{AttMean}}$) achieves **93.84%** Top-1 consistency. This indicates that in the vast majority of cases, the decision sentence's attribution mass is concentrated on the source chunk corresponding to the option chosen by the model. In other words, the generation-time tracing mechanism recovers the model's own selected option as the most supported source unit.

Interpretation. This result provides strong evidence that OmniTrace captures attribution signals aligned with the model's internal decision pathway. If attribution were dominated by post-hoc noise or unrelated spans, the consistency rate would approach chance level (25% for four-way multiple choice). Instead, the observed 93.84% rate suggests that generation-time token tracing, combined with span-level aggregation and filtering, faithfully identifies the textual source corresponding to the model's final answer statement.

C. Scoring functions and baselines

C.1. Choice of Base Attribution Methods

OmniTrace is designed as a framework that converts arbitrary token-level attribution signals into generation-time, span-level source explanations. In principle, many attribution methods could serve as the underlying token-level signal. In this work, we adopt three representative methods: **AttMean**, **RawAtt [9]**, and **AttGrad [14]**. These methods were selected because they satisfy two practical requirements for generation-time attribution in decoder-only multimodal LLMs: (1) they operate directly on model-internal signals available during decoding (e.g., attention weights or gradients), and (2) they can be computed efficiently without requiring additional supervision or model modifications.

Attention-based attribution. We include two attention-based methods as lightweight and widely used attribution signals. **RawAtt** uses the mean attention weights from the final Transformer layer as token attribution scores [9]. **AttMean** aggregates attention weights across all layers and heads, providing a more stable signal by averaging attention distributions throughout the network. These methods are particularly suitable for generation-time attribution because attention weights are readily available during decoding via standard model interfaces (e.g., `output_attentions=True`) and can be computed without additional backward passes.

Gradient-based attribution. To complement attention-based signals, we also include **AttGrad**, which combines gradient information with attention maps to estimate token influence [14]. Gradient-based methods provide a different attribution perspective by measuring how changes in input representations affect the generation probability of output tokens. Although gradient computation requires an additional backward pass, it remains feasible for decoder-only models when attribution is computed per decoding step.

Methods not included. Several other attribution methods have been proposed in the literature, including Layer-wise Relevance Propagation (LRP) [10], attention rollout [9] and AttCAT [16]. However, these approaches are less directly compatible with generation-time attribution in decoder-only multimodal LLMs. Many were originally developed for classification settings with fixed prediction targets, whereas generation requires attribution to be computed online for each decoding step. Some methods also rely on full attention matrices over all token pairs or repeated forward passes over the entire sequence, which becomes computationally expensive for long-context generation and multimodal inputs given the scales of the state-of-the-art Omni-modal LLMs.

Our goal is not to exhaustively benchmark all attribution techniques, but to demonstrate that OmniTrace functions as a *plug-and-play framework*. By showing consistent performance improvements across diverse base signals—including attention-only and gradient-based variants—we illustrate that the framework generalizes across attribution mechanisms rather than relying on a specific scoring heuristic.

Computational considerations. Gradient-based attribution ($\text{OT}_{\text{AttGrads}}$) is memory-intensive for long audio and video inputs, and thus results are not reported for those settings (†). Nevertheless, in visual tasks where gradients are tractable, $\text{OT}_{\text{AttGrads}}$ remains competitive, further supporting the signal-agnostic nature of the framework.

C.2. Baselines

Here we discussed the implementaton of our baselines in Table 2.

C.2.1. Self-attribution

After model generation is complete, the same base model is prompted using the prompts from Appendix B.1 to attribute each of it's generation to the sources provided. For visual tasks we supply chunked visual and text elements from source. For audio and video tasks we supply the raw audio/video files.

C.2.2. Embedding-based heuristics

After model generation is complete, each of the chunk in model generation gets embedded either using the base model's processor ($\texttt{Embed}_{\texttt{processor}}$) or using openai/clip-vit-large-patch14 ($\texttt{Embed}_{\texttt{CLIP}}$). For visual tasks, each of the chunk from source is also embedded the same way, and we compute the cosine-similarity of the embeddings and keep those above a threshold. We did a hyperparameter sweep on the threshold and kept 0.25 which yield the highest performance.

C.2.3. Random

This is a post-hoc method. Implementation is similar to Embedding-based heuristics except that the matching is random instead of similarity-based.

Embedding-based baselines are not applicable to audio and video attribution tasks (marked $\times$ in Table 2). These baselines rely on computing similarity between generated spans and discrete source embeddings. However, in temporal attribution settings, source labels correspond to continuous timestamp intervals rather than semantically self-contained discrete units. Embedding similarity cannot be meaningfully defined over arbitrary time bins without an external segmentation model, which would introduce additional supervision and confound comparison. Therefore, embedding-based matching is inherently unsuitable for continuous timestamp attribution, whereas OmniTrace operates directly over token-level causal signals and does not require discrete semantic chunk embeddings.

D. Model details

D.1. Base model

Here are the generation methods for the open-sourced models.

For **Qwen2.5-Omni**, we implemented 7B versions following the official repository: https://github.com/QwenLM/Qwen2.5-Omni.

For **MiniCPM-o-4.5**, we implemented 9B versions following the official repository: https://github.com/OpenBMB/MiniCPM-o.

D.2. ASR model

For ASR models, we implement the Paraformer as the default for audio summarization. And we include two other state-of-the-art variants in our ablation study.

For Paraformer, we implemented the model from huggingface https://huggingface.co/funasr/paraformer-zh following the official guide: https://github.com/modelscope/FunASR.

For Whisper, we implement the the model from huggingface https://huggingface.co/openai/whisper-large-v3 following the official guide: https://github.com/openai/whisper.

For Scribe v2, we implement following the offical repo:

https://elevenlabs.io/blog/introducing-scribe-v2.

E. Data Release

We will publicly release a comprehensive code base that includes the OmniTrace implementation with different scoring functions.

We would also release the test set. The licensing terms for the artifacts will follow those set by the respective dataset creators, as referenced in this work, while the curated artifacts will be provided under the MIT License. Additionally, our release will include standardized evaluation protocols, and evaluation scripts to facilitate rigorous assessment. The entire project will be open-sourced, ensuring free access for research and academic purposes.